



CONTINUOUS BIOPROCESSING

Continuous bioprocessing control and connectivity

TS

**CONNECT
COLLABORATE
ACCELERATE™**

Contents

1.0	Introduction	7
2.0	Continuous process connectivity and control	8
	2.1 Process flow considerations	8
	2.2 Unit operation interface considerations	13
	2.3 Intermediate vessel considerations	14
	2.4 Control schemes	16
	2.4.1 Identifying and applying consolidated control schemes	17
	2.5 Process orchestration: local vs global control	22
	2.6 Application of conserved control schemes between unit operations	23
3.0	Bioburden control and other risk considerations	26
4.0	Conclusion	28
	Appendix	30
	Acronyms	42
	References	43

List of figures

Figure 1: Distillation of a continuous interconnected bioprocess into three control strategies	6
Figure 2: Block flow diagram of example process layouts	9
Figure 3: Qualitative visualization of the potential product stream parameters over time (t), characterized by volumetric flow (F), concentration (c), and resulting product mass flow (m)	12
Figure 4: Matrix of potential product stream interfaces between unit operations in an integrated continuous process based on example scenarios of product stream volumetric flow (F, L/h), concentration (c, g/L), and resulting mass flow (m, g/h). Considering the interfaces between unit operations, A, B, C, and D refer to the output stream of the preceding unit operation and 1, 2, 3, and 4 refer to the required input stream parameters for the subsequent unit operation	14
Figure 5: Conserved control scheme 1: control approach for unit operations not requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors	18
Figure 6: Conserved control scheme 2: control approach for unit operations optionally requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors	19
Figure 7: Conserved Control Schema 3: control approach for unit operations requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors	20
Figure 8: Summary to show how the three control schemes could fit into a standard mAb continuous process. Numbers 1, 2 and 3 are the control schemes in Figures 5, 6 and 7	21
Figure 9: Differentiation of local (blue) and global (orange) process orchestration layers. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors	22
Figure 10: Assignment of unit operation linkages to each conserved control scheme	29

List of tables

Table 1: LpH VI: dual tank unit operation, with a control scheme assignment, potential critical process parameters/key process parameters, ranges and putative alarm scenarios	24
Table 2: Non-exhaustive list of other considerations for the final control strategy for a continuous bioprocess	27

About BioPhorum

BioPhorum's mission is to create environments where the global biopharmaceutical and device industry can collaborate and accelerate its rate of progress, for the benefit of all.

Since its inception in 2004, BioPhorum has become the open and trusted environment where senior leaders of the biopharmaceutical industry come together to openly share and discuss the emerging trends and challenges facing their industry.

Growing from an end-user group in 2008, BioPhorum's membership now comprises top leaders and subject matter experts from global biopharmaceutical manufacturers and suppliers, working in both long-established and new Phorums. They articulate the industry's technology roadmap, define the supply partner practices of the future, and develop and adopt best practices in drug substance, fill finish, process development and manufacturing IT.

In each of these Phorums, BioPhorum facilitators bring leaders together to create future visions, mobilize teams of experts on the opportunities, create partnerships that enable change and provide the quickest route to implementation, so that the industry shares, learns and builds the best solutions together.

Authors

Asahi Kasei Bioprocess

Donald Gibbs

CRB

Alejandro Kaiser

Emerson

Bob Lenich

Exyte (formerly M + W)

Carl Carlson

FUJIFILM Diosynth Biotechnologies

Charles Heise

GSK

Ashley Reeder

Janssen

Paul Randolph

Lonza

Sandeep Hadpe

Cytiva

Mark Schofield

Rockwell Automation

Bruce Kane

Siemens

Pamela Docherty

Thermo Fisher Pharma Services

Kristina Pleitt

BioPhorum

Graeme Moody

Contributors

Asahi Kasei Bioprocess

Julie Kozaili

Merck

Michael Phillips

Andrea Ruggiero

Veena Rao

Pascal Valax

Novo Nordisk

Matthias Wiendahl

PM Group

Charles Heffernan

Sartorius Stedim

Martin Lobedann

Schneider Electric

Vicki Pearson

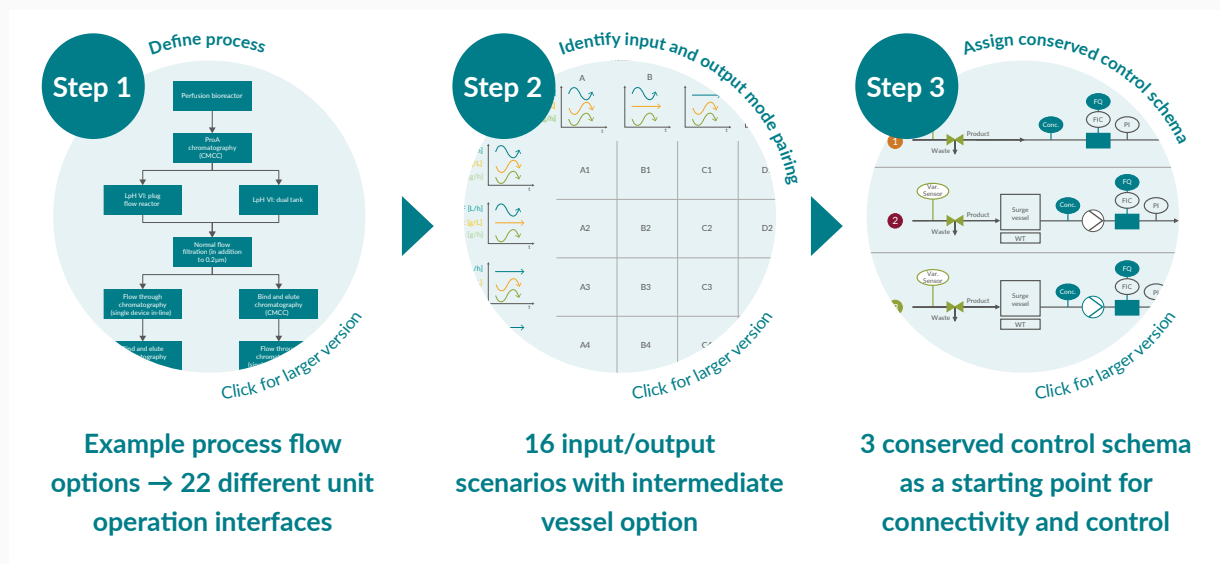
Executive summary

Cost-effective, rapid and flexible biopharmaceutical production is an industry-wide challenge. Process intensification through continuous manufacturing has proved successful in other industries and is a promising area of investigation for researchers, vendors and biomanufacturers. However, business and technical barriers, both real as well as perceived, have hindered the widespread adoption of continuous bioprocessing¹. The challenge of developing an end-to-end, continuous bioprocess has been addressed by several companies with a range of solutions¹. More recently, a common framework for integrated and continuous biomanufacturing has emerged: perfusion processes linked to multi-column capture chromatography predominate, with some processes being integrated and continuous from the bioreactor to the final concentration step². Moving beyond batch operation, linking process steps together and interdependence between the two require new and more complicated control strategies. This increase in complexity is cited as a hurdle to adoption. Through collaboration and a multi-faceted approach, BioPhorum aims to rationalize and simplify the product handoff between unit operations.

This paper aims to align and standardize the current industry approach to control of a continuous monoclonal antibody (mAb) bioprocess^{3,4}. It builds on the BioPhorum risk assessment template that defined an integrated continuous bioprocessing control strategy⁵. The first aim is to understand the flow of product in an interconnected process. This is used to bring clarity to the continuous bioprocess end state, reduce the perceived complexity of continuous bioprocessing control strategies and lower the implementation barrier for continuous bioprocessing. The ultimate goal is to increase access to biopharmaceuticals by reducing cost and time to market as well as improving control of product quality¹.

This paper distils complex interactions between bioprocess unit operations into three basic control building blocks (Figure 1). Considerations are included for the implementation and control of a continuous process which mitigate potential process risks. The paper defines a control approach based on the specific requirements of a given process using a conserved approach to linking the outputs and inputs between unit operations.

Figure 1: Distillation of a continuous interconnected bioprocess into three control strategies



Target readership:

- Bioprocess engineers, process control personnel and automation engineers who will design and implement continuous processes for the manufacture of biotherapeutic molecules. This is for understanding connectivity, possible control strategies and potential process risk considerations
- Technical leaders in both biomanufacturing and contract organizations that are considering the implementation of continuous processing capabilities at their sites. This document supports understanding some basic elements for continuous process control, identification of control scheme for connecting unit operations, as well as consideration of some of the potential risks and possible mitigation strategies.



“Continuous bioprocessing represents the next evolution in cost-effective manufacture of certain therapies and will improve facility utilization, product quality and ultimately patient access. I’m very excited to have played a part in articulating a control framework that will reduce barriers to entry for biomanufacturers who are seeking to develop continuous biomanufacturing processes. I am actively working on projects that will leverage these methodologies and it will be exciting to see them in practice!”

Charles Heffernan, Director of Advanced Therapies and Biologics, PM Group

1.0

Introduction

At its heart, continuous processing potentially simplifies manufacturing by operating at a steady state. Continuous processes have already been developed for several small molecule therapeutics, but the complexities of biologics have slowed the adoption of continuous bioprocessing. A key challenge identified in a previous risk assessment⁵ was the necessity to link unit operations to operate a continuous bioprocess. This paper defines a standard template approach for developing and implementing a continuous bioprocessing control strategy. The template provides a visual and technical aid for linking common unit operations in the continuous biomanufacturing process, and highlights considerations when selecting control schemes. This paper does not define unit operations or their order within a process. This paper outlines a means of assessing the unique design needs of a continuous bioprocess to establish a connected product flow across all integrated unit operations. This approach includes three steps (Figure 1):

Step 1

Define process: determine the order of unit operations and identify process linkages.

Step 2

Identify input and output mode pairing: define the material flow in and out of each unit operation pairing as varying, constant concentration and/or volumetric flow.

Step 3

Assign conserved control scheme: select from three conserved control schemes as for connectivity of unit operations.

Although the number of unique permutations of process flow at the interface of two operations may initially appear complex, Section 3.4 describes how three conserved control schemes can be utilized to manage all scenarios. These simple control strategies can be implemented via a combination of local and global automation layers to manage the product and information flow between modules.



“Our experience with continuous manufacturing is that it does not need to be unnecessarily complicated. The product handoff between unit operations is important and seemingly quite varied, but with more consideration can be consolidated to three simple strategies. This provides clarity to the interconnectivity and rationalizes the control strategies.”

Mark Schofield, Cytiva

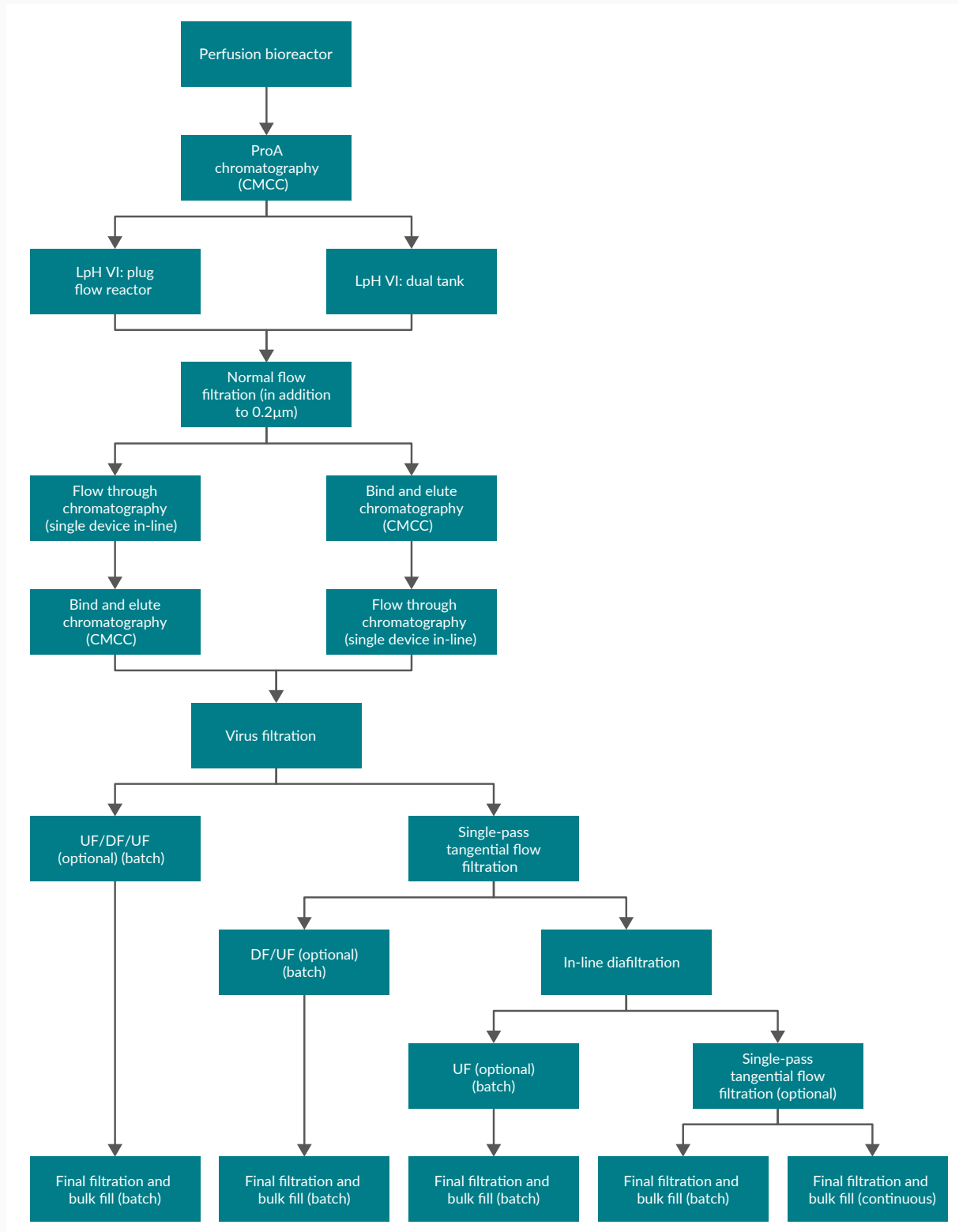
2.0

Continuous process connectivity and control

2.1 Process flow considerations

The unit operations and regulatory batch definitions required to produce an mAb are well defined^{2, 7, 8, 9}. This paper identifies options to transfer the batch process into a continuous or semi-continuous process as described in *A risk-based blueprint for process control of continuous bioprocessing*⁵. Each of the unit operation functions is maintained in the transition from batch to continuous. There is often a range of solutions at each step to achieve process intensification at the unit operation level and for the overall process. Additional detail on batch definitions is out of scope; a definition is given in the continuous bioprocessing roadmap document¹. However, the way a batch is defined in the process (e.g. time, volume or product mass) may influence the control strategy.

Figure 2: Block flow diagram of example process layouts



To create a control strategy, assumptions were made based on the process flow steps in Figure 2:

- The perfusion bioreactor is taken to include both the cell retention device and reactor vessel as a single entity generating a constant volumetric flow
- The Protein A (ProA) capture step is via continuous multi-column chromatography (CMCC) that enables continuous loading but has sequential intermittent elution
- The low-pH viral inactivation (LpH VI) step is usually one of two formats: plug flow reactor, where material continuously moves through a flow path with a specified residence time, or dual tank, where material is collected and held for the requisite time in sub-batches before forward processing
- Normal flow filtration is performed with a backup filter that can be brought online as the first filter fouls
- Polishing chromatography is performed in multi-column bind and elute or single device in-line (SDIL) flow through modes executed continuously (without interconnection of columns for loading), where the columns are sized to achieve a cadence that maintains continuous operation. SDIL refers to one column or membrane in the main product stream at a time. There may be multiple devices operated in parallel or sequentially, but not loaded in series.
- Viral filtration is operated in continuous flow mode. Another filter can be brought in-line to allow integrity testing of the initial filter
- Concentration and buffer exchange in a continuous process can be achieved through continuous single pass concentration and buffer exchange, pooling of material and batch ultrafiltration (UF) and diafiltration (DF) or a hybrid solution
- The final filtration of the drug substance is typically a batchwise operation. There is an opportunity to explore continuous filtration and bulk drug substance (BDS) fill, to generate drug substance lots for disposition for a prescribed total batch time, volume or concentration.

There are multiple options for the creation of BDS. Figure 2 shows, for a process with nine to 11 unit operations, there are eight to 10 interfaces where product exits one unit operation and enters another. However, considering the diversity of processes there are many more potential linkages (Figure 2 shows 22 different linkages). Understanding product handoff from one unit operation to another is key in developing an integrated process and could impact the choice of process intensification solution at unit operation level.



“Key to demonstrating a continuous process in a state of control is the demonstration of a functionally closed system. The impact to the product zone from external influences must be managed through risk assessments and secondary process controls.”

Carl Carlson,
formerly Exyte

Figure 3 shows how each linkage between two individual unit operations can be characterized by the product concentration (c), volumetric flow rate (F) and the resulting product mass flow (m). This is a visual representation, used to indicate consistency or change over time and is not a reflection of exact processing conditions in terms of direction, magnitude or rate for each parameter, or relative to other parameters. Five scenarios are identified:

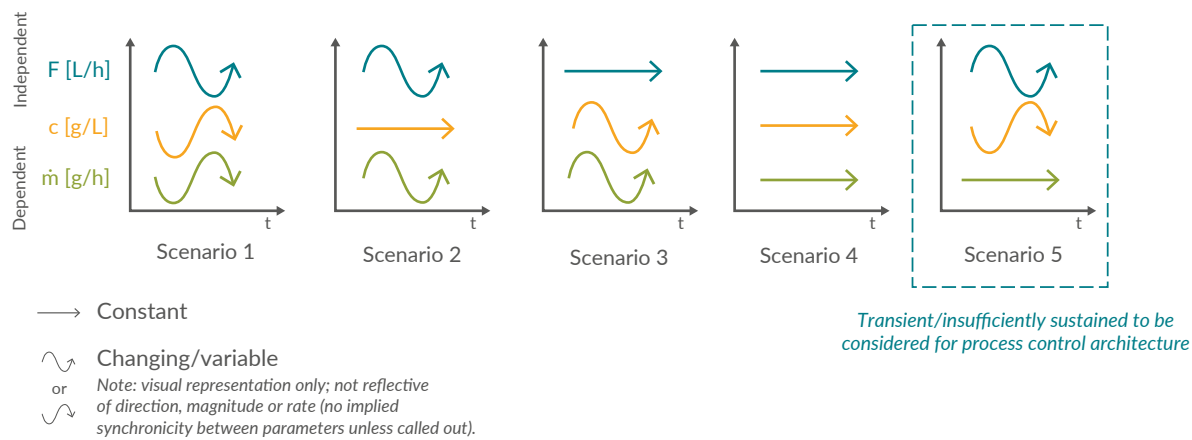
- **Scenario 1:** when the volumetric flow rate and concentration vary in a non-synchronic manner, the resulting product mass flow rate changes over time. This scenario assumes that flow rate and concentration vary inversely and proportionally in a 1:1 ratio. Other types of non-synchronic variation are possible.
Examples: normal operation of a perfusion bioreactor or the protein A eluate stream
- **Scenario 2:** constant concentration and varying volumetric flow rate, resulting in a variable product mass flow.
Examples: single-pass tangential flow filtration (SPTFF) or filtration of a homogeneous feed with either a fouling filter or variable flow rate to regulate product mass flow
- **Scenario 3:** when the volumetric flow rate remains constant and the concentration varies, a varying product mass flow rate results.
Example: a continuous viral inactivation
- **Scenario 4:** behavior probably only achievable for short durations of time, where both constant flow rate and concentration result in a constant product mass flow.
Examples: during a flow through chromatography or non-fouling filter step

- **Scenario 5:** a theoretical stream composition, where volumetric flow and concentration are inversely proportional and synchronized to achieve a constant product mass flow. Scenario 5 is not sufficiently sustained over the duration of the unit operation to be considered a unique condition and is therefore typically considered a subset of Scenario 1.

The varying volumetric flow rate scenarios include zero flow, periodic flows or unit operation pauses. The varying concentration scenarios include zero product concentration (e.g. due to a product chase step), so that the product mass flow rate is also zero at the same point in time in the downstream process.

The qualitative visualization of the potential product stream parameters over time are shown in Figure 3. F and c are considered independent parameters as both can and are commonly adjusted during processing. $\dot{m}$ is considered a dependent parameter as it is a result of F and c . Five example situations were considered: Example I) Variable flow and concentration resulting in varying product mass flow. II) Varying flow and constant concentration resulting in varying product mass flow. III) Constant flow, variable concentration resulting in variable product mass flow. IV) Constant flow and concentration resulting in constant product mass flow. These are qualitative representations of the expected process input and output scenarios within an integrated process. Example V is insufficiently sustained to be considered for process control architecture.

Figure 3: Qualitative visualization of the potential product stream parameters over time (t), characterized by volumetric flow (F), concentration (c) and resulting product mass flow ($\dot{m}$)



“Process intensification by interconnecting multiple unit operations will have a significant impact on process economics due to reduced floor space requirements and plant productivity due to simultaneous execution of purification steps.”

Dr Martin Lobedann,
Sartorius

2.2 Unit operation interface considerations

A review of the 22 potential linkages between unit operations (Figure 2) identified five possible input or output scenarios (Figure 3). Scenarios 1 to 4 are considered distinct. Scenario 5 can be considered a subset of Scenario 1, and therefore does not necessitate its own classification of control.

Material is supplied to a biomanufacturing unit operation with either varying volumetric flow (Scenarios 1 and 2) or constant volumetric flow (Scenarios 3 and 4). Both options can be further subdivided depending on whether the concentration is varying (Scenarios 1 and 3) or constant (Scenarios 2 and 4). Similarly, the product mass flow output from a biomanufacturing unit operation can then be categorized as either constant or varying.

Flow control strategies in unit operations, which usually determine the output flow rates, can be operated in different modes. The simplest option uses the output

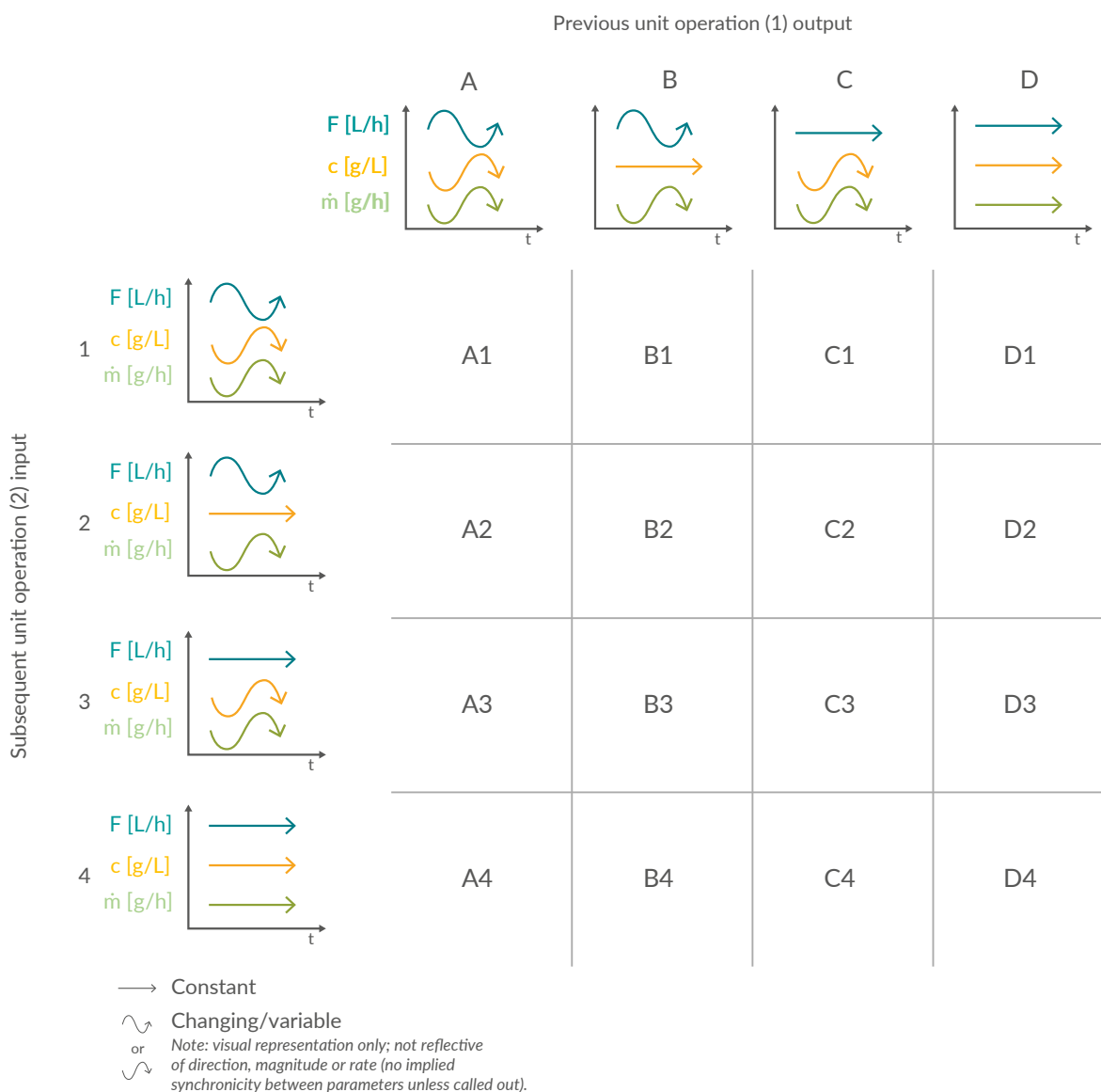
flow rate from the preceding unit operation to determine the input flow rate of the next. The most sophisticated option is the steady state flow, where the fill level of the intermediate tanks is kept close to a pre-defined setpoint with a proportional integral derivative controller, allowing for small adaptations in flow rate. Here, only minor differences in the flow rate will occur. The third option is to control the flow rates by certain pre-defined thresholds in fill level or recipe stages with a state-based controller, where the flow is adjusted to pre-defined values, e.g. low, medium and high flow rate. The fourth option is the on/off control, where the flow rate is either running at high value or zero. The transition from one output scenario to an input scenario between the unit operations forms the basis of the underlying building blocks of the liquid and product mass flows through a continuous process. Figure 4 illustrates 16 unit operation interface permutations arising from the four input and output scenarios.



“When nesting multiple control loops, one must remember to give internal control strategies appropriately responsive algorithms, on a timescale order of magnitude. Failure to have appropriately responsive algorithms can result in unwelcome process cycling.”

Bruce Kane, PE, Industry Consultant
Rockwell Automation

Figure 4: Matrix of potential product stream interfaces between unit operations in an integrated continuous process based on example scenarios of product stream volumetric flow (F , L/h), concentration (c , g/L) and resulting mass flow ($\dot{m}$, g/h). Considering the interfaces between unit operations, A, B, C and D refer to the output stream of the preceding unit operation and 1, 2, 3 and 4 refer to the required input stream parameters for the subsequent unit operation



2.3 Intermediate vessel considerations

There are advantages to including an intermediate vessel between unit operations in a continuous process⁷. These include normalization of flow, pressure equilibration, minimization of perturbation propagation, dampening of concentration gradients, removal of entrained air, sampling of homogeneous material and potential for

in-line solution adjustment (e.g. pH, conductivity, dilution, etc.) or provision of an in-process hold point⁸. Inclusion of an intermediate vessel also enables relative decoupling of unit operations to aid process control, process deviation management and equipment failure mitigation. If the surge vessels are large in comparison to the flow rate, they can provide time for maintenance and replacement. However, additional vessels also increase

the overall system cost and require periodic cleaning and potential sanitization. Automation complexity is increased through additional control elements, including sensors to maintain fill level or pressure within specific ranges, hold up volume, local residence time and the residence time distribution. Therefore, the size of each intermediate vessel should be considered based on minimizing the impact of the additional volume, while ensuring the vessel operates above a minimum level to prevent air entrapment and ensure sufficient mixing, where applicable, and below a maximum to prevent integrity loss and residence time broadening. In closed-system operation, dead zones in the process flow should be taken into account when switching between surge tanks for tank cleaning or replacement. In some instances, where application of dual surge tanks and process systems is required (i.e. two identical systems operated in parallel to provide redundancy or necessary downtime for changeover, cleaning or maintenance), these switching points provide opportunities for cleaning and instrument changeover.

Categorizing unit operation inputs and outputs into 16 distinct unit operation interfaces provides a framework for assigning intermediate vessels. Surge vessels may be necessary at specific linkages to mitigate differences between the unit operation cadences, and to enable multiple periodic effluents to be accumulated prior to forward processing. In some instances, the unit operation can accommodate **volumetric changes**, e.g. tank-based viral inactivation, whereas others

cannot, e.g. single column bind/elute chromatography, and require incorporation of an intermediate vessel. Selection of the specific intermediate vessel will be driven by the process risks identified in the risk assessment⁵, leading to potential reassignment of a unit operation linkage into a different control scheme and choice of a surge or cycle vessel. Simplistically, these are either surge or pooling vessels. Adopting the National Institute for Innovation in Manufacturing Biopharmaceuticals (NIIMBL) categorization of intermediate vessels as being a small surge tank, a cycle surge tank or a batch pool tank, specific input and output combinations can be identified³. The small surge tank, used to accommodate small flow fluctuations and pressure differences, could be incorporated into process linkages where the inputs and outputs match one of the unit operations. This can mitigate for the input/output requirements of the subsequent/previous unit operation. The cycle surge tank, used to accumulate material from sub-cycles to smooth the output, could be deployed where the output is periodic and feeds into an input with a constant cadence. The batch pool tank could be deployed at any point in the process to pool material for homogeneity, to provide an in-process hold point or potentially to define single drug substance lots. The bases of the control strategy approach can be identified, through assignment of the flow scenarios and control scheme to the linkages in a continuous biomanufacturing process. Typical linkages in a continuous bioprocess are discussed in Section 2.4.



“While continuous biopharmaceutical manufacturing meets the pressing need for cost-effective, responsive manufacturing solutions that adapt to fluctuating market demand, implementation is often hindered by the perceived complexity of the operations. By distilling the control scheme down to its fundamental components, this paper provides a tangible guide for teams seeking to design and implement continuous manufacturing environments.”

Ashley Reeder, Investigator at GSK

2.4 Control schemes

This section outlines the rationale for simplifying a complex process flow diagram (Figure 2) to a core set of three schemes that describe the control between unit operation linkages. Integrated continuous bioprocesses vary depending on the molecule, manufacturer and facility, however, the interfaces between the unit operations are conserved. This paper is focused on the input and output scenarios considered, based on industry experience, the most likely options for continuous mAb processing. The 16 possible permutations of four inputs and four outputs can be consolidated into three control schemes (Figures 5, 6 and 7). Consideration of existing material cadence into the unit operation and desired material cadence, informs the decision for inclusion of an intermediate vessel between two given unit operations, as described in Section 2.3. Linkages between two apparently identical unit operations may require different control schemes in different processes, e.g. viral filtration, where different filter types may require different flow and pressure characteristics to maintain the same viral clearance capabilities.

The control schemes in Figures 5, 6 and 7 include CQA-guided elements for forward processing selection associated with the output of the upstream unit. If product quality falls outside of specification, as monitored by a process sensor, the associated valve (after unit operation 1) is utilized to divert material to waste until material attributes can be brought back within range. For simplicity, this assumes that diverted product is not recirculated back to the surge tank or otherwise quarantined for reprocessing. The small loss of yield incurred via this strategy is acknowledged and accepted. Flow control elements associated with the inlet of the downstream unit are included as they are required to satisfy process constraints.

A brief description of each of the three control schemes is included with examples to provide a starting point and guidance on each scheme that can be customized to meet specific process needs. The examples are not exhaustive but provide a frame of reference.



“Connectivity between the different operations is a focal point in continuous processing, setting it apart from batch manufacturing. Managing the variability in concentration, volume and mass flow rate is critical for the success of the operation, in which in-process analytical technology is key.”

Alejandro Kaiser, PE,
Senior Process Engineer at CRB

2.4.1 Identifying and applying consolidated control schemes

Identifying and applying the most suitable consolidated control scheme for controlling the passing of material between steps in a continuous process is an exercise that requires detailed understanding of the details, assumptions and critical parameters surrounding each unit operation and linkage in the specific process. There is no one-size-fits-all control recommendation, but one of the three identified control schemes will provide an excellent place to build from. The recommended approach for applying this to a specific continuous process for implementation is as follows in these steps:

1. Identify the unit operation that precedes the linkage to a subsequent unit operation. For example, virus filtration (VF) is the preceding unit that links to single pass tangential flow filtration (SPTFF) in a hypothetical process flow (refer to Figure 2 for unit operation sequence).
2. List the operational characteristics and assumptions that govern that first unit operation leading to the identification of a specific output from that operation. VF would be the first unit operation per the example given in step 1. This is one of four combinations for both output velocity (flow rate) and product concentration (mass flow).
 - a. Options are as listed in Figure 4, and labeled with the following **letter** designation:
 1. Constant mass flow and constant flow rate—**D**
 2. Variable mass flow and constant flow rate—**C**
 3. Constant mass flow and variable flow rate—**B**
 4. Variable mass flow and variable flow rate—**A**
3. Decide which of the four best characterizes the output of the initial unit operation and assign it the appropriate letter label.
4. Identify the unit operation with the linkage that succeeds the identified initial unit (as in the example given per step 1, SPTFF succeeds VF).
5. List the operational characteristics and assumptions that govern that subsequent unit operation leading to the identification of a specific input type for that unit operation which is required for a successful process outcome. Again, SPTFF would be the subsequent unit operation per the example SPTFF given in step 1. This is one of four combinations for both output velocity (flow rate) and product concentration (mass flow).
 - a. Options are as listed in Figure 4, and labeled with the following **numeric** designation:
 1. Constant mass flow and constant flow rate—**4**
 2. Variable mass flow and constant flow rate—**3**
 3. Constant mass flow and variable flow rate—**2**
 4. Variable mass flow and variable flow rate—**1**
6. The final output of this application method yields an alphanumeric code. In this example **D4** was identified based on the output effluent and input requirements associated with these two specific unit operations. This becomes the linkage type identifier. From this, match the identifier to the recommended consolidated control scheme 1, 2 or 3 (Figures 5, 6 and 7). In this example, **D4** is matched to **consolidated control scheme 2** which gives a good starting point to develop a control strategy for that linkage.

This is a starting point and should not be considered prescriptive. Criticality of operations and risks identified may lead to adoption of a different control scheme (to the one recommended for the linkage type identifier) that better reflects the tolerance to process variability and to potential failure of the unit operation.

Note: Appendix 1 includes a linkage identifier to control scheme table and specific process examples for how this application method can be used to recommend consolidated control schemes for controlling unit operation linking in a continuous process.

Consolidated control scheme 1

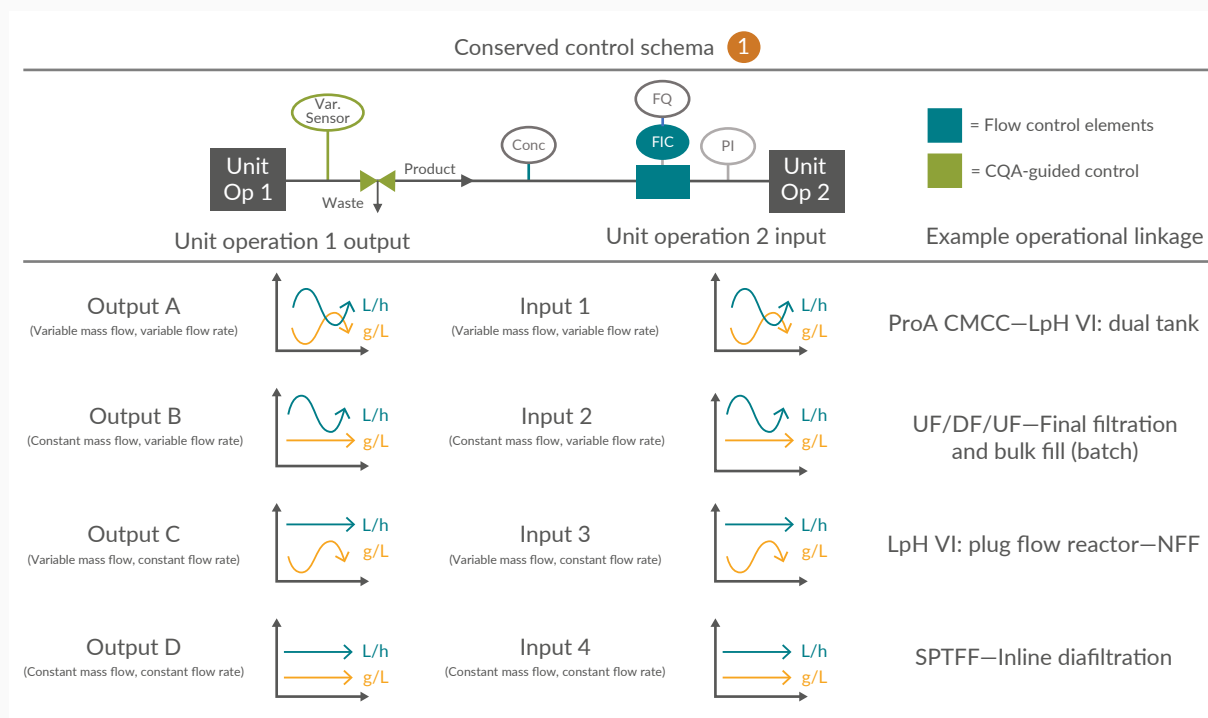
Control scheme 1 outlines a process linkage in which there is *no intermediate surge vessel*. This is unique to this control scheme. It is anticipated that in-line concentration monitoring through, for example, ultraviolet (UV) and totalizing flow control will be provided to regulate product mass and/or volumetric load across the subsequent unit operation. Alternatives to UV tracking may be employed depending on the product, accuracy and resolution of the detection system and the technology maturity. However, for simplicity, UV tracking is used as a generic descriptor for real-time quantitative detection.

Taking the NFF to FT chromatography operational linkage C3 as an example, the UV sensor may be used to model protein concentration and the flow totalizer used to calculate total load across the FT chromatography

column. In this way, these elements are utilized to monitor protein loading of a flow through column and to redirect the feed stream to a second chromatography column upon reaching a load ratio limit.

In certain circumstances the decision may be made to employ control scheme 2, which includes an intermediate vessel, in place of control scheme 1 to satisfy the risk profile for a given operation. Such a surge vessel serves to dampen the impact of process variability and to decrease the likelihood of a downstream process pause, should there be an interruption to processing upstream. Such interruptions to input flow, for example, are to be avoided by first intent across viral filters to mitigate risk for potential viral breakthrough. Alternatively, driven by a need to define a lot/batch, the option to use control scheme 3 may be implemented.

Figure 5: Conserved control scheme 1: control approach for unit operations not requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors



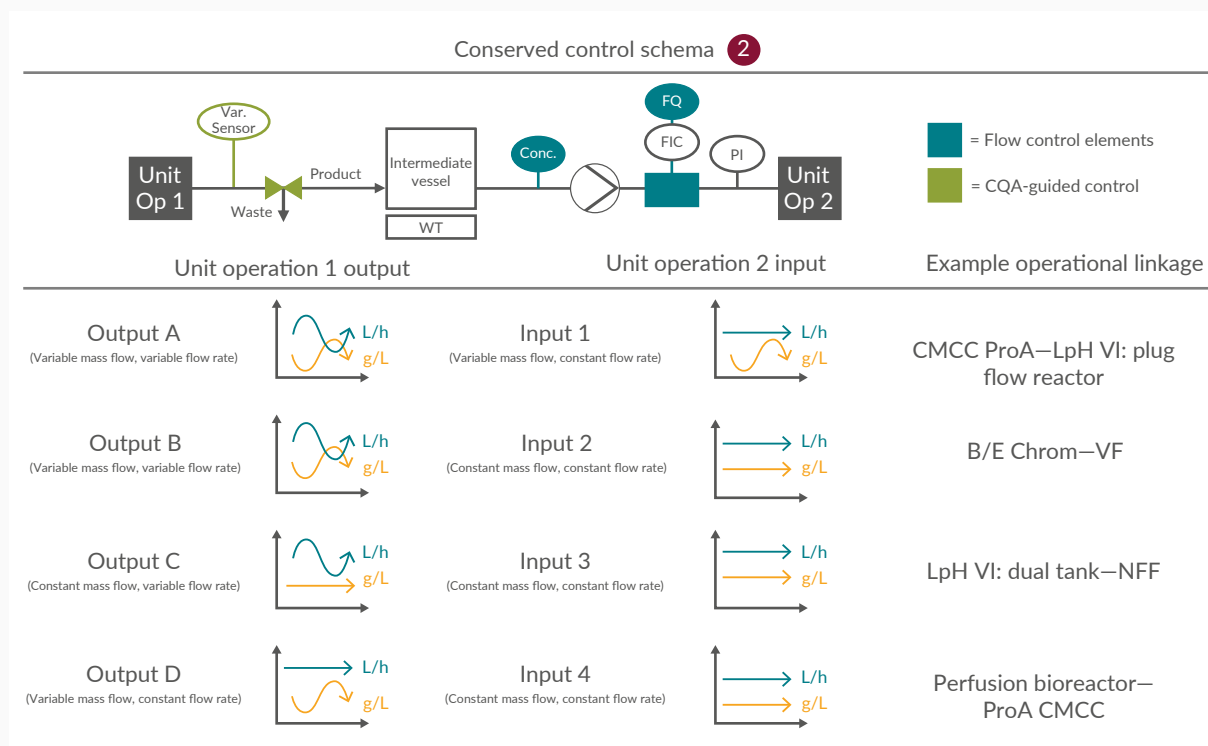
Consolidated control scheme 2

Control scheme 2 is one in which a small surge or cycle surge tank is required. This is due to a mismatch between required output and input material cadence and vice versa. This scheme incorporates a UV monitor and totalizing control support to manage mass loading across the subsequent unit operation. Due to the introduction of an intermediate vessel, an additional pump is required to manage the flow rate from the surge vessel into the subsequent unit operation.

Taking the VF to SPTFF operational linkage C2 as an example, the flow meter controls the SPTFF unit operation and, depending on the process tolerances, product mass fluctuations and pressure profiles will drive the flow rate to target a specific concentration. This requires a surge vessel to mitigate for the constant flow output from the VF step.

Control scheme 2, e.g. D4, may be implemented even when the operational linkage does not demand an intermediate surge vessel, as inclusion can provide additional process tolerance and continuity.

Figure 6: Conserved control scheme 2: control approach for unit operations optionally requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors

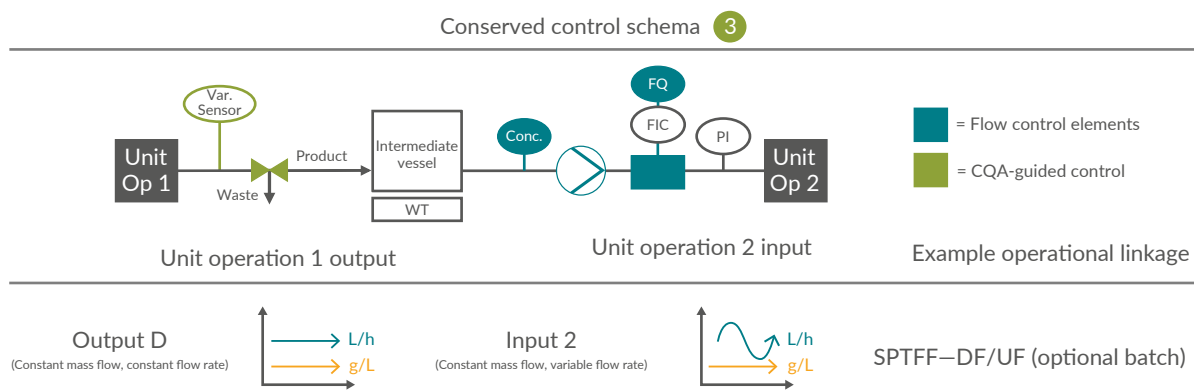


Consolidated control scheme 3

Control scheme 3 requires an intermediate vessel due to the mismatch between the desired material cadences of output and input unit operations. Unlike in control

scheme 2, this linkage necessitates a batch pool tank to facilitate the combination and homogenization of output material over a longer time interval up to inclusion of the entire batch, prior to forward processing, as illustrated by B4 LpH VI: dual tank to NFF.

Figure 7: Conserved Control Schema 3: control approach for unit operations requiring a surge vessel. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors



Eleven inlet/outlet cadence permutations were assigned to a scheme above from the representative flow diagram in Figure 2. Linkage A1 was assigned to control scheme 1 as alignment of the expected flow rates was assumed to be synchronous. If the A1 linkage was assumed to be asynchronous then control scheme 2 would be appropriate. Linkages A2, B1, B3, C1, C2, D1 and D3 were not assigned as no practical processing

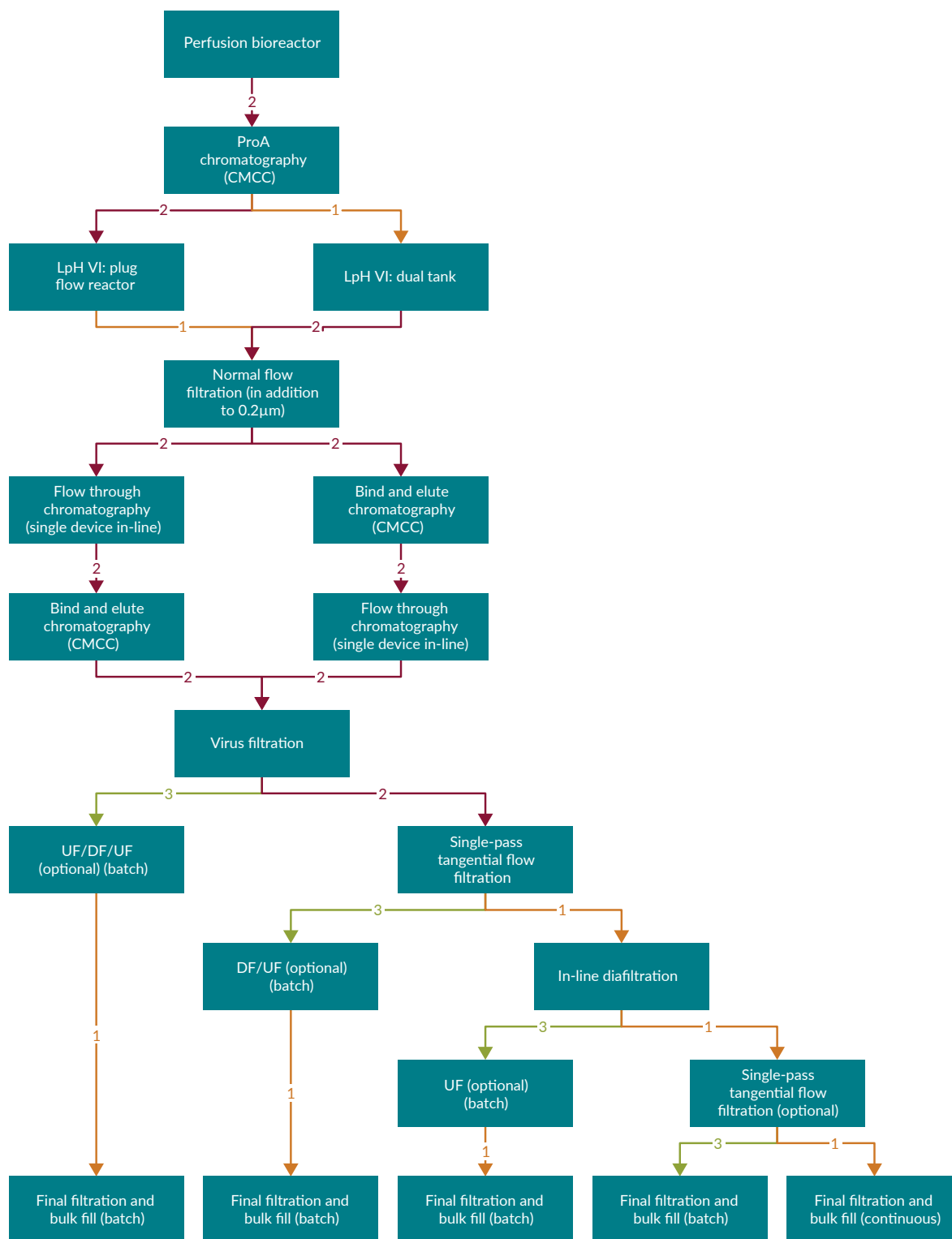
scenario was identified to which these could be applied. However, all except D3 would be controlled using control scheme 2 or potentially control scheme 3 if feed stream homogenization was required. Linkage D3 would be assigned to control scheme 1. Figure 8 shows an end-to-end process assignment for the process flow represented in Figure 2. It is stressed that this is a conditional assignment and not prescriptive of all expected processes.



“Continuous processing offers sustainable bioprocessing since it results in improved operational efficiencies, reduced cost of goods sold and improved product quality over batch. Integrating operation with minimum surge and reliable automation and control strategy is key for a successful continuous operation.”

Dr Sandeep Hadpe,
Lonza

Figure 8: Summary to show how the three control schemes could fit into a standard mAb continuous process. Numbers 1, 2 and 3 are the control schemes in Figures 5, 6 and 7

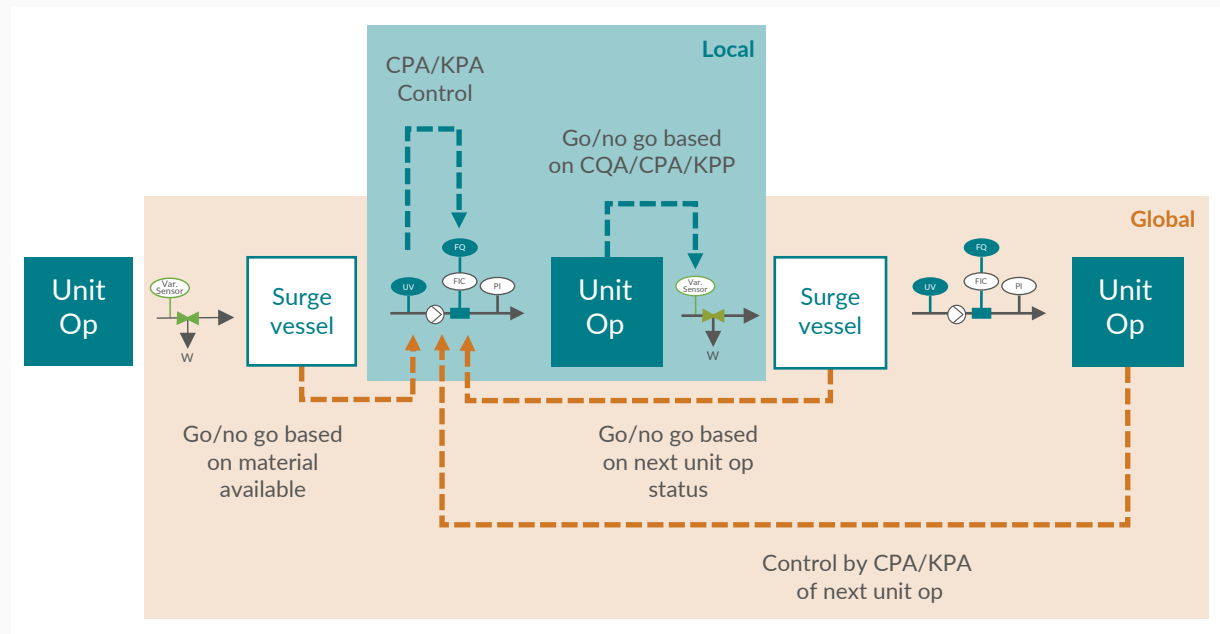


2.5 Process orchestration: local vs global control

The consolidated control scheme defines the means to coordinate product handoff between two unit operations and the requirement for a surge vessel. Process orchestration layers (POLs) are assigned to these schemes to characterize the control requirements at each unit operation transition. In a connected process, control of every material transition is typically governed by the requirements of the in-line monitors within the specific control scheme element and downstream unit operation (local). Whereas the global

POL considers the process as a whole and the linkages between the process steps upstream and downstream of the control schema element. This hierarchical information flow, either directly from in-line sensors or from algorithms, simplifies the identification of the control requirements at each step. The output from the risk assessment⁵ identifies whether these elements are controlled at a unit operation level or whether there may be broader global linkages. In Figure 9, local control responses (blue) operate within the boundaries of the process step and global control responses (orange) allow responses from the wider process.

Figure 9: Differentiation of local (blue) and global (orange) process orchestration layers. FQ flow totalizer, FIC flow meter, PI pressure indicator, UV product concentration measurement and Var. Sensor unit operation specific process sensors



One option for orchestrating an integrated and continuous downstream process is to use a global automation layer, which is connected to all unit operations (top-down approach). Each unit operation only communicates with the orchestration layer, supplying values and receiving new setpoints. Such a setup enables straightforward integration for multiple unit operations while only requiring a single type of interface. The orchestration layer manages the distribution of information, meaning data generated on one system can be used for feedback or feedforward control by other unit operations. This layer therefore provides an optimal location for the implementation of control loops to produce a more sophisticated, 'intelligent' layer, allowing the actual unit operations to stay relatively simple. For this scheme to function, the unit operations must have the ability to act on pause and resume signals, and receive setpoints.

Another option for orchestrating an integrated downstream process relies on direct system-to-system communication (bottom-up approach). This requires no additional automation layer, which can reduce costs but will likely lead to increased variance within the interfaces of different unit operations. Each unit operation has a

sophisticated control scheme to synchronize with its neighbor. An important consideration for this type of orchestration is the difficulty in transmitting data to secondary connections. This is a result of the inherent design of the system, which only allows exchanging information via direct connections.

Selection of either option described above will be dependent on the infrastructure, validation plan and operating strategy versus the cost associated with implementing one or the other automation approach.

2.6 Application of conserved control schemes between unit operations

Appendix 1 exemplifies the process and rationale for selecting conserved control schemes for the unit operations in Figure 2. Table 1 provides an illustrative example of the LpH VI: dual tank unit operation, with a control scheme assignment, potential critical process parameters/key process parameters, ranges and putative alarm scenarios. It is envisaged that these examples provide a basis for demonstrating the concepts described in this document for establishing the process for developing a control strategy for a continuous process.

Table 1: LpH VI: dual tank unit operation, with a control scheme assignment, potential critical process parameters/key process parameters, ranges and putative alarm scenarios

Unit ops linkage under consideration	Control scheme	Controlled parameter	Parameter manipulated	Example parameter range	Control hierarchy	Potential alarm-action setup
Protein A CMCC to LpH VI: dual tank	Control scheme 1	Volume in VI tank 1	ProA flow rate	X-Y mL/min or X-Y minute column residence time	Global	HiHi: Divert Protein A eluate to waste Hi: Increase periodicity between Protein A elution events by decreasing Protein A flow rate/increasing Protein A residence time Lo: Decrease periodicity between Protein A elution events by increasing Protein A flow rate/decreasing Protein A residence time
	Control scheme 1	Volume in VI tank 1	Level control setpoint	Minimum level control setpoint	Local	LoLo: Pause acid addition to low pH vessel until sufficient volume accumulated
	Control scheme 1	LpH residence time	ProA flow rate	X-Y mL/min or X-Y minute column residence time	Global	Hi: Decrease periodicity between Protein A elution events by increasing Protein A flow rate Lo: Increase periodicity between Protein A elution events by decreasing Protein A flow rate to ensure maximum residence hold times met
	Control scheme 1	LpH residence time	N/A	N/A	Local	HiHi: Divert titrated product to waste LoLo: Divert titrate product to waste due to incomplete viral inactivation
	Control scheme 1	pH	Volumetric addition of acid/base	Low pH target 3.5 +/- 0.1	Local	Hi: Increase volume of acid added to tank Lo: Add base to neutralize to target as needed (uncommon practice) LoLo: Divert out of range product to waste
	Control scheme 1	Product concentration in tank	Level control setpoint (# elution events to collect before titration) Volumetric addition of acid	Eluate A280 conc.	Local	Hi: Pause acid titration operation until additional eluates collected Hi: Increase amount of acid titrant to reach low target pH (controlled by acid pump PID loop) Lo: Decrease amount of acid titrant added to reach low pH target

The process monitoring requirements for a continuous process are not necessarily more onerous than for a batch process. Critically, process monitoring needs to meet the requirements of the risk assessment to allow demonstration of process control⁹. The frequency of sampling/monitoring and any additional monitoring needs will be identified in the risk assessment over an equivalent batch process. However, processing information from the in-process sensors and detection method outputs will differ as PAT feedback and feedforward information is incorporated into the appropriate control scheme at each linkage point. These control loops can be either local or global (Figure 10) and are assigned against the critical quality attribute (CQA), critical process attribute (CPA) and key process attribute

(KPA) outputs from the risk assessment for each unit operation. The specific control logic deployed can range from empirically determined normal operating ranges and setpoints to advanced digital twin-based process control models and algorithms. Whichever approach is adopted, the complexity of a highly controlled continuous bioprocess can be developed based on a set of common foundation blocks and principles. Building the global control on the foundations of the local control loops with PAT feedback and feedforward logic will structure the complex control strategy in a comprehensible format. In turn, this will simplify the approach to the validation activities that need to be undertaken for licensing continuously manufactured biologics.

As the control strategy is developed from the risk assessment, PAT considerations should be mitigated against. Measurement equipment should be evaluated for instrumentation signal stability, accuracy and repeatability for the lifecycle of the run, the limits of quantification and speed of measurement for a given detection method and its precision against pharmacopeia standards. The feasibility of direct measurement against indirect (e.g. calculated or inferred) measurement and any regulatory concerns over the latter must also be considered. If an in-process change of equipment or measurement device is required, then the impact on the control strategy must be evaluated. One advantage of the incorporation of PAT is the enabling of continuous process validation, which has been recognized as an alternative approach to validate a control strategy through trending and evaluating process parameters and attributes¹⁰. This enables an evidence-based approach to define appropriate invasive sampling and at-line measurement frequencies that minimize the need to compromise the integrity of the process boundary. Finally, the control strategy for a continuous process must deliver product that meets the specified quality characteristics of the pharmaceutical product (QTPP).

Within the continuous manufacturing paradigm, 'material traceability' and 'lot/batch genealogy' should be used to describe the process for mapping how incoming raw material lots are distributed to final product lots. A material tracing mechanism ensures that raw material found to have a quality issue during or after production can be thoroughly investigated and steps taken to guarantee product safety. In continuous processes, raw materials are constantly converted by, or interacting with, the product flow stream. A change in an input as well as a material conversion due to a unit operation will take time to propagate through the system and be seen in the final output. Moreover, the stream itself is not a first-in-first-out process but includes significant mixing across all the unit operations and surge tanks. Additionally, the longer material remains in the continuous line, the more mixing and potential impact of a material quality problem during production could affect large production volumes. Continuous lines include material extraction points to allow for removal of material at key points in the operation. Material tracking/traceability calculations must incorporate these considerations into overall material tracking.



“Demonstrating control of a complex, continuously operating process is challenging without identifying the underlying common building blocks and themes. Simplifying the detection, traceability and control elements based on a consensus view will de-risk adoption of continuous processing for large molecule biomanufacturing.”

Dr Charles Heise,
FUJIFILM Diosynth Biotechnologies

3.0

Bioburden control and other risk considerations

Bioburden control is critical for maintaining low-bioburden levels in continuous bioprocessing, with different strategies required for success at each unit operation. Strategies for maintaining low bioburden levels used in batch processes may not be sufficient for longer processing times¹¹. Successful control strategies will include process design for prevention of microbial incursion, maintenance of system integrity, monitoring for detection of bioburden and pre-defined escalation and response protocols. Each of these elements must be evaluated using a risk assessment approach (for example Failure Mode Effects and Analysis (FMEA)). Additionally, validation and testing should be implemented thoughtfully to address potential risks identified in the risk assessments and should support the full duration of process operations. This includes choice of materials and consumable components, understanding the impact of their use through the entire duration of a continuous process, along with a plan for potential failure during operation. Robustness of materials and equipment is critical to maintenance of system integrity in a continuous operation.

As noted in Section 4.1 (process monitoring), sampling frequency for bioburden/endotoxin will be identified in the risk assessment and incorporated into the process control strategy. This is to support the establishment and maintenance of a stable, controlled state and accurate reflection of system status at any given time¹². The frequency must also support the bracketing and segregation of impacted product in instances of process failure. It is expected that the response to process failure be clearly defined prior to operations and that the disposition of product follows that guidance. Where a continuous process adopts a continuous monitoring strategy, sampling frequencies may be re-evaluated and reduced in response to continuous process validation.

Alternatively, mature processes with embedded process analytical technology (PAT) may allow detection of specific process drift scenarios, which could action pre-defined sampling and bioburden testing responses rather than utilizing a regular bioburden sampling routine. Regulatory expectations for biological manufacturing processes must be able to produce a sterile drug product with a high degree of assurance¹³.

Table 2 lists considerations for microbial control in implementation of a continuous process. The list is not intended to be exhaustive but gives five basic categories and corresponding ideas that should be considered in system design, process controls and operation.

Table 2: Non-exhaustive list of other considerations for the final control strategy for a continuous bioprocess

Category	Unit ops linkage under consideration
Prevention of ingress	Build a closed system with controlled single-use sterile/irradiated materials and consumables that meet process requirements from reliable vendors
	Sterile filter non-sterile solutions at point of use
	Aseptic connect and tube weld all connection points
	Build change-out strategies for equipment/component replacement including redundancy at high-risk connections into lines, tube sets and other addition/removal points
Maintenance of system integrity	Design cleaning cycles/washes with effective bactericides for in-process sanitization of system routes (e.g. chromatography columns)
	Strategically implement sterilizing grade filters into the system design with replacement strategies in place (e.g. chromatography column effluents)
Process monitoring	Select sampling sensitivity, sample locations (e.g. pre-filtration) and sample frequencies that support process control for the duration of the entire operation
	Build effective strategies for data analysis and trending to accurately reflect the status of the process at defined points in time
	Use rapid testing to assure timely access to process information
	Use in-line monitoring where possible
	Build monitoring strategies into process control and process automation
Escalation and response	Set pre-defined response actions for out-of-specification data and drifting trends
	Create procedures for assessment, bracketing and segregation for out-of-control product and materials
	Build capability for effective root cause analysis
	Set guidelines for continued use of chromatography resins or other cycled materials in the continuous process
Validation	Consider using media runs or testing in situ as operation validation
	Consider increased and targeted non-routine sampling in qualification runs
	Perform testing on materials, components and equipment to assess robustness over maximum process duration and to understand potential wear and breakdown issues along with defining system maintenance
	Any process intended hold times in process intermediate vessels at specific unit operations where batch processing is employed should be microbial hold validated



“Implementation of a continuous manufacturing process for a biotherapeutic molecule can seem especially daunting given the number of unit operations, inputs/outputs and parameters that need to be set up and controlled. Understanding the connectivity, using basic control schema, along with thoughtful consideration and mitigation of potential process risks can lead to more successful outcomes in execution of these processes at manufacturing scale.”

Don Gibbs, Senior Manager at Asahi Kasei Bioprocess America

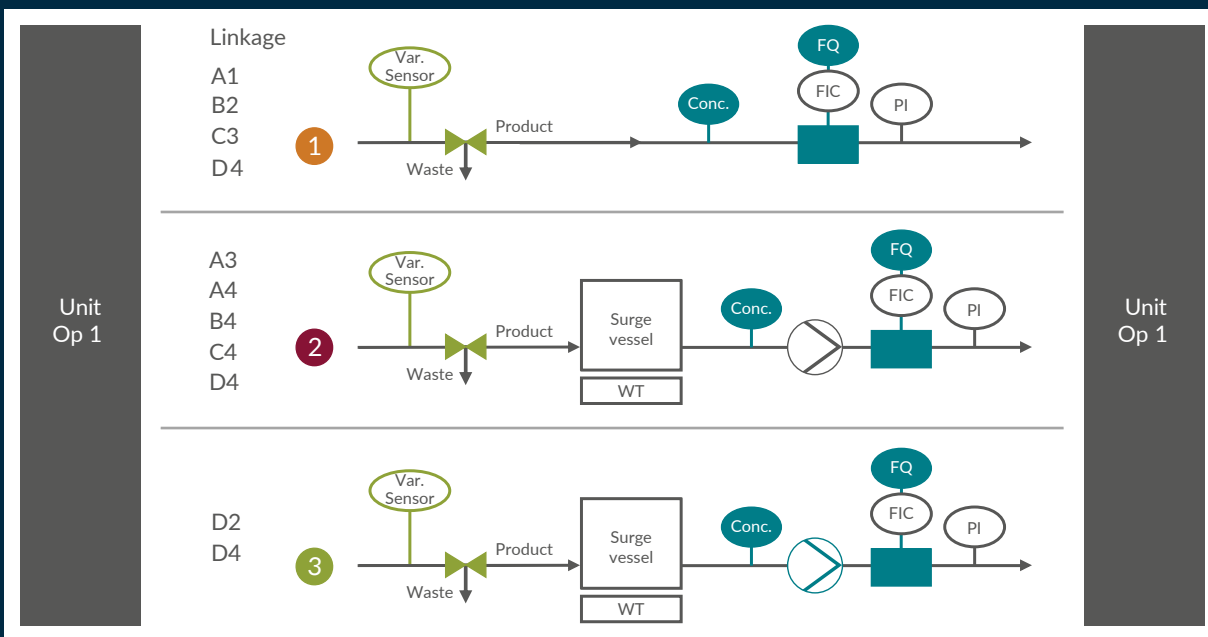
Conclusion

This paper reduces the product handoff between one unit operation and the next to just three possible control schemes. This simplifies the description of connections between process steps considerably. By starting with an example mAb platform process, 22 discrete connections were identified. These were rationalized to 16 possible combinations based on the nature of the volumetric mass flow out of a unit operation and the volumetric mass flow into the subsequent operation. With just three control schemes relevant for the interconnection of process steps, interconnections can become conceptually and physically modular. This is important as it drives simplicity in building the process. It creates the possibility to have a standard set of valving, surge tanks and PAT employed for each control scheme. This simplifies the number of skids that must be ordered and may help reduce the inventory required to handle deviations or process failures. Additionally, by using the same interconnection strategy between multiple operations it simplifies the control strategy: the same control strategy can be cut and pasted between all connections that use the same scheme.

The process for assigning a conserved control scheme between the unit operations of a standard continuous biomanufacturing process is as follows (Figure 1):

1. Define the continuous process unit operations or steps
2. Identify input/output pairings and intermediate vessel requirements
3. Assign the conserved control schema to the identified pairings (Figure 10).

Figure 10: Assignment of unit operation linkages to each conserved control scheme



This allows building of an appropriate control strategy, based on key process parameters, CQAs, environmental controls, sampling, data analysis/trending and event response in conjunction with the identified conserved control schemes as outlined in this paper.

This rationalization of the interconnections will help simplify development and operation of a continuous automated bioprocess. By simplifying in this way, the intention is to lower the barriers to adoption of continuous processing. The implementation of continuous processing still offers potential to improve product quality through improved product consistency by operating at a steady state and to improve patient access by reducing manufacturing cost through a more intensive use of equipment and consumables.



“Although operationally beneficial, increasing automation and eliminating downtime between unit operations can feel like a risky, complicated aspect of continuous bioprocessing (especially for those more removed from development and implementation). Here we show how focusing on a few fundamentals and following a simple, systematic exercise makes the process control concept and strategy approachable to even those who are newer to integrated, continuous bioprocessing.”

Kristina Pleitt, Senior Manager at Thermo Fisher Scientific

Appendix

Distilling the potential complexity of interactions between unit operations and linkages into specific assignments for control strategies requires a detailed understanding of how each step is being performed, the sources of variability and process risks. Included here are examples of how to create the alphanumeric code discussed in the paper and assign a corresponding consolidated control scheme. Figure 2 provides the unit operation roadmap for these example assessments resulting in the actual control scheme assignment recommendations. This appendix provides guidance on how to perform the work, based on assessment of each linkage in a continuous process for manufacture of a monoclonal antibody. The recommended results will not accurately reflect all unique process requirements. Each assessment in this appendix looks at the output of the first unit operation to identify the constant, variable or intermittent nature of both the flow and product concentration. This output assessment is then matched to the subsequent unit operation's specific input requirements and variability tolerances leading to the recommended control scheme best suited to control the product transfer of that linkage. The notes section also includes assumptions and risks for the completed assessment giving insight into the logic for the specific recommendation. The assessments are intended to provide clear, concrete examples on how to apply the contents of this paper to an actual continuous process for implementation in a manufacturing facility.

Definitions

Input/output assignments:

- **Constant:** no notable change over time; minor fluctuations may occur (e.g. result of small process or measurement fluctuations)
- **Variable:** significant change over time due to variation in processing conditions (e.g. pump setting)
- **Intermittent:** on/off change over time.

Vessels:

- **Cycle surge:** one cycle is collected, then intermediate volume is further processed
- **Small surge:** mean residence time <2 hours
- **Batch pool tank:** whole volume of the batch is collected and mixed, then further processed.

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Perfusion bioreactor	Volumetric flow rate (L/h)	Constant	Small surge	Volumetric flow rate (L/h)	Constant	ProA chromatography (CMCC)
		Product concentration (g/L)	Variable		Product concentration (g/L)	Constant	
Input/output pairing	C				4		
Conserved control scheme	2						
Notes	• The volumetric flow rate of the perfusate is variable prior to steady-state operation (typically ≤5 days after perfusion starts). The perfusate during the lead-up to steady state is not forward processed. After steady state is reached, the volumetric output of the perfusion bioreactor is assumed to be constant (i.e. bioreactor feed and bleed rates are also constant) • The perfusion bioreactor product concentration output is considered variable in response to intermittent cell bleeding • It is assumed that the ProA CMCC is designed to accommodate a constant volumetric feed flow rate (i.e. loading flow is not interrupted and the loading step duration limited). It is possible that the unit operation is designed with a variable load flow rate or the load phase is intermittent • The product concentration into the ProA CMCC is assumed to be constant as consistent product concentration aids in consistent cycle time management for CMCC. However, as for the bioreactor output, the product concentration in the perfusate may vary, but little to no impact is anticipated for variations in load concentration for affinity chromatography • A small surge vessel is assumed to be most commonly used between the perfusion and ProA CMCC to accommodate small fluctuations in flow rate (e.g. changes in bioreactor bleed rate), equipment or maintenance activities (e.g. column guard filter change out), create a break point for sterility between upstream and downstream processing, and enable sampling prior to purification.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	ProA chromatography (CMCC)	Volumetric flow rate (L/h)	Variable/intermittent	Small surge	Volumetric flow rate (L/h)	Constant	LpH VI: plug flow reactor
		Product concentration (g/L)	Variable		Product concentration (g/L)	Variable	
Input/output pairing	A				3		
Conserved control scheme	2						
Notes	• The ProA CMCC output is classified as variable or intermittent as product elutions would occur periodically throughout processing (i.e. volumetric flow rate out of the system would be intermittent (on/off)) • The product concentration is variable as elution peak is eluted from the column in a non-uniform concentration with respect to time • Low pH viral inactivation using a plug flow reactor requires constant volumetric flow rate into the system to ensure uniform residence time • The product concentration into the system is variable, assuming a buffer flush is periodically incorporated between products. During times of product introduction, it is assumed that a uniform product concentration would be targeted for ease of validation • A small surge vessel is selected to accommodate the mismatch in output and input requirements (i.e. intermittent output to a constant input of volumetric flow rate and create a uniform product concentration between unit operations).						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	ProA chromatography (CMCC)	Volumetric flow rate (L/h)	Variable/intermittent	None	Volumetric flow rate (L/h)	Variable/intermittent	LpH VI: dual tank
		Product concentration (g/L)	Variable		Product concentration (g/L)	Variable	
Input/output pairing	A				1		
Conserved control scheme	1						
Notes	• The ProA CMCC output was classified as variable or intermittent as product elutions would occur periodically throughout processing (i.e. volumetric flow rate out of the system would be intermittent) • The product concentration is variable as elution peak is eluted from the column at a non-uniform concentration with respect to time • LpH VI using a dual tank setup has intermittent volumetric flow rate into each of the tanks • This assumes that the intermittent flow rates for ProA CMCC and LpH VI are linked and therefore do not require a surge vessel • It is assumed the ProA column elution(s) would be collected in one of the LpH VI tanks and, therefore, the inlet concentration is classified as variable • The tanks in the dual tank system setup act as a cycle surge vessel and an additional vessel is not typically required. It is possible that the ProA CMCC elution cadence and/or elution volume may lead to an alternative setup, however, this is considered uncommon • An intermediate vessel is not required to accommodate the unit operation interfaces in terms of flow. However, a small surge vessel may be required to account for system pressure depending on cadence.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	LpH VI: plug flow reactor	Volumetric flow rate (L/h)	Constant	None	Volumetric flow rate (L/h)	Constant	Normal flow filtration (in addition to 0.2µm)	
		Product concentration (g/L)	Constant		Product concentration (g/L)	Variable		
Input/output pairing	C				3			
Conserved control scheme	1							
Notes	• LpH VI using a plug flow reactor requires constant volumetric flow rate through the system to ensure uniform residence time • The product concentration out of the system would be variable assuming a buffer flush is periodically incorporated between product. During times of product introduction, it is assumed that a uniform product concentration would be targeted for ease of validation • It is assumed constant volumetric flow rate would be targeted for the normal flow filtration to avoid pauses that may impact filter performance. However, some normal flow filtration may not require constant volumetric flow operation • It is assumed product concentration into the normal flow filter would be variable if buffer flushes are periodically incorporated between products. However, normal flow filtration could be operated in a manner where the product concentration into the system is constant • An intermediate vessel is not required to accommodate the unit operation interfaces in terms of flow. However, a small surge vessel may be required to account for system pressure depending on overall setup.							

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	LpH VI: dual tank	Volumetric flow rate (L/h)	Variable/intermittent	Cycle surge	Volumetric flow rate (L/h)	Constant	Normal flow filtration (in addition to 0.2 μm)	
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant		
Input/output pairing	B				4			
Conserved control scheme	2							
Notes	• LpH VI using a dual tank setup has intermittent volumetric flow rate out of each of the tanks/the system • As the product is mixed in the tanks of the system, the product concentration out to a low pH dual tank setup is constant • It is assumed constant volumetric flow rate would be targeted for the normal flow filtration to avoid pauses that may impact filter performance. However, some normal flow filtration may not require constant volumetric flow operation • It is assumed product concentration into the normal flow filter would not vary significantly even if buffer flushes were periodically incorporated between product. As such, we assume normal flow filtration could be operated in a manner where the product concentration into the system was effectively constant • The tanks in the dual tank system setup act as a cycle surge vessel and an additional vessel is not typically required. It is assumed that buffer flushes would occur to accommodate the mismatch in volumetric flow requirements between the unit operations.							

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	Normal flow filtration (in addition to 0.2µm)	Volumetric flow rate (L/h)	Constant	Small surge	Volumetric flow rate (L/h)	Constant	Flow through chromatography (single device in-line)	
		Product concentration (g/L)	Constant or variable		Product concentration (g/L)	Constant		
Input/output pairing	D or C				4			
Conserved control scheme	2							
Notes	- It is assumed constant volumetric flow rate would be targeted for the normal flow filtration to avoid pauses that may impact filter performance. However, some normal flow filtration may not require constant volumetric flow operation - It is assumed product concentration out of the normal flow filter would be constant if using dual tank LpH VI or variable if using plug flow VI - The volumetric flow rate into the flow through chromatography unit operation was assumed to be constant to maintain a uniform residence time for product. This assumes that the column/device has sufficient capacity (device scale or application in parallel) to avoid discontinuous load - It is assumed product concentration into the FT chromatography would not vary significantly even if buffer flushes of normal flow filter (NFF) are periodically incorporated between products. As such, it is assumed normal flow filtration could be operated in a manner where the product concentration into the system was effectively constant - A small surge vessel is assumed to be most commonly used between normal flow filtration and FT chromatography irrespective of VI approach.							

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Normal flow filtration (in addition to 0.2µm)	Volumetric flow rate (L/h)	Constant	Small surge*	Volumetric flow rate (L/h)	Constant	Bind and elute chromatography (CMCC)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme	2*						
Notes	- It is assumed constant volumetric flow rate would be targeted for the normal flow filtration to avoid pauses that may impact filter performance. However, some normal flow filtration may not require constant volumetric flow operation - It is assumed product concentration out of the normal flow filter would be operated in a manner where the product concentration into the system was constant - It is assumed that the bind and elute CMCC was designed to accommodate a constant volumetric feed flow rate (i.e. loading flow is not interrupted or loading step duration limited). It is possible that the unit operation is designed with a variable load flow rate or the load phase is intermittent (single device). This may result in a different intermediate vessel or cycle surge tank and control schema selection (D2) - The product concentration into the bind and elute CMCC was assumed to be constant as consistent product concentration aids in consistent cycle time management for CMCC. *Although D4 falls within the remit of control schema 1, a small surge vessel is assumed to be most commonly used between normal flow filtration and bind and elute CMCC to dampen concentration/pH/conductivity variations or adjustments, accommodate equipment or maintenance activities (e.g. column guard filter change out), minimize impact of flow rate differences and aid in pressure drop.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Flow through chromatography (single device in-line)	Volumetric flow rate (L/h)	Constant	Small surge*	Volumetric flow rate (L/h)	Constant	Bind and elute chromatography (CMCC)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme	2*						
Notes	• The volumetric flow rate out of the flow through chromatography unit operation is assumed to be constant with the column/device having sufficient capacity (device scale or application in parallel) to avoid discontinuous flow • It is assumed product concentration out of the FT chromatography would not vary significantly even if buffer flushes are periodically incorporated • It is assumed that the bind and elute CMCC is designed to accommodate a constant volumetric feed flow rate (i.e. loading flow is not interrupted or loading step duration limited). It is possible that the unit operation is designed with a variable load flow rate or the load phase is intermittent. This may result in a different intermediate vessel and control scheme selection • The product concentration into the bind and elute CMCC is assumed to be constant as consistent product concentration aids in consistent cycle time management for CMCC. *Although D4 falls within the remit of control scheme 1, a small surge vessel is assumed to be most commonly used between flow through and bind and elute chromatography. While an intermediate vessel is not required to accommodate the unit operation interfaces in terms of flow, a small surge vessel may be required to account for system pressure depending on overall setup.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Bind and elute chromatography (CMCC)	Volumetric flow rate (L/h)	Variable/intermittent	Cycle surge	Volumetric flow rate (L/h)	Constant	Flow through chromatography (single device in-line)
		Product concentration (g/L)	Variable		Product concentration (g/L)	Variable	
Input/output pairing	A				3		
Conserved control scheme	2						
Notes	• The bind and elute CMCC output flow is classified as variable or intermittent as product elutions occur periodically throughout processing (i.e. volumetric flow rate out of the system would be on/off) • The product concentration is variable as elution peak is eluted from the column at a non-uniform concentration with respect to time. • The volumetric flow rate into the flow through chromatography unit operation is assumed to be constant to maintain a uniform residence time for product. However, depending on the number of devices (in parallel), the input may be variable/intermittent if the column/membrane is periodically taken offline for cleaning. This may require a different intermediate vessel and control scheme • The product concentration into the flow through chromatography step would be variable (e.g. column needs to be taken offline for cleaning and the product is chased with buffer to improve recovery) • A cycle surge vessel is selected to accommodate the mismatch in output and input requirements (i.e. intermittent output to a constant input of volumetric flow rate and create a uniform product concentration between unit operations).						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Bind and elute chromatography (CMCC)	Volumetric flow rate (L/h)	Variable/intermittent	Cycle surge	Volumetric flow rate (L/h)	Constant	Virus filtration
		Product concentration (g/L)	Variable		Product concentration (g/L)	Constant	
Input/output pairing	A				4		
Conserved control scheme	2						
Notes	• The bind and elute CMCC output flow is classified as variable or intermittent as product elutions occur periodically throughout processing (i.e. volumetric flow rate out of the system would be on/off) • The product concentration is variable as elution peak is eluted from the column at a non-uniform concentration with respect to time • The volumetric flow rate into the virus filter is assumed to be constant to avoid process pauses/differential pressure changes across the filter • The product concentration into the virus filter is assumed to be constant. While buffer flushing may be periodically incorporated if having to switch between virus filters (to improve recovery), published data support extended use of the filters and it is assumed that change out is an infrequent occurrence compared to other process filters and devices • A cycle surge vessel is selected to accommodate the mismatch in output and input requirements (i.e. intermittent output to a constant input of volumetric flow rate and create a uniform product concentration between unit operations).						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Flow through chromatography (single device in-line)	Volumetric flow rate (L/h)	Constant	Small surge*	Volumetric flow rate (L/h)	Constant	Virus filtration
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme	2*						
Notes	• The volumetric flow rate out of the flow through chromatography unit operation is assumed to be constant. However, depending on the number of devices (in parallel), and the input strategy for the flow through chromatography operation, the output may be variable/intermittent with pauses as a new device comes online • The product concentration out of the flow through chromatography step would be variable (e.g. column needs to be taken offline for cleaning and the product is chased with buffer to improve recovery) • The volumetric flow rate into the virus filter is assumed to be constant to avoid process pauses/differential pressure changes across the filter • The product concentration into the virus filter is assumed to be constant. While buffer flushing may be periodically incorporated if having to switch between virus filters (to improve recovery), published data support extended use of the filters and it is assumed that change out is an infrequent occurrence compared to other process filters and devices. Additionally, potential impact in the product concentration to the downstream unit operations may not justify recovery of the system hold-up volume *Although D4 falls within the remit of control scheme 1, a small surge vessel is assumed to be most commonly used between flow through chromatography and virus filter to dampen concentration/pH/conductivity variations or adjustments and as a precautionary measure as the performance of the virus filtration step is critical for safety/viral validation claims.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Virus filtration	Volumetric flow rate (L/h)	Constant	Batch pool*	Volumetric flow rate (L/h)	Variable/intermittent	UF/DF/UF—optional (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				2		
Conserved control scheme				3*			
Notes	• The volumetric flow rate out of the virus filter is assumed to be constant • The product concentration out of the virus filter is assumed to be constant. While buffer flushing may be periodically incorporated if having to switch between virus filters (to improve recovery and mitigate risk in the process by incorporating periodic integrity testing of the virus filter), published data support extended use of the filters and it is assumed that change out is an infrequent occurrence compared to other process filters and devices • Variable/intermittent volumetric flow rate into the batch UF/DF/UF operation is assumed (i.e. specified amount of product collected and then batch processed). Initial concentration may occur as product is introduced to the system. *A batch pool tank was assumed to be used for lot definition.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Virus filtration	Volumetric flow rate (L/h)	Constant	Small surge*	Volumetric flow rate (L/h)	Constant	Single-pass tangential flow filtration
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme				2*			
Notes	• The volumetric flow rate out of the virus filter is assumed to be constant • The product concentration out of the virus filter is assumed to be constant. While buffer flushing may be periodically incorporated if having to switch between virus filters (to improve recovery and mitigate risk in the process by incorporating periodic integrity testing of the virus filter), published data supports extended use of the filters and it is assumed that change out is an infrequent occurrence compared to other process filters and device • Constant volumetric flow rate into the single-pass tangential flow filter is assumed if the concentration out of the virus filter is constant *Although D4 falls within the remit of control scheme 1, a small surge vessel is assumed to be most commonly used between the viral filter and SP-TFF. While an intermediate vessel is not required to accommodate the unit operation interfaces in terms of flow, a small surge vessel may be required to account for system pressure or flow rate depending on overall setup.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	UF/DF/UF—optional (batch)	Volumetric flow rate (L/h)	Variable/intermittent	None	Volumetric flow rate (L/h)	Variable	Final filtration and bulk fill (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	B				2		
Conserved control scheme	1						
Notes	• The volumetric flow rate out of the batch UF/DF/UF operation is variable/intermittent as it is performed in batch mode • The product concentration out of the batch UF/DF/UF operation is constant • The volumetric flow rate into the batch final filtration and bulk fill may be variable/intermittent depending on the setup • The product concentration into the final filtration and bulk fill is constant • No intermediate vessel is assumed as the material was batch pooled in the prior unit operation and the retentate vessel acts as the feed for the bulk fill.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Single-pass tangential flow filtration	Volumetric flow rate (L/h)	Constant	Batch pool*	Volumetric flow rate (L/h)	Variable/intermittent	DF/UF—optional (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				2		
Conserved control scheme	3*						
Notes	• Constant volumetric flow rate out of the single-pass tangential flow filter is assumed based on if the inlet concentration is constant • Constant product concentration out of the single-pass tangential flow filter is assumed • Variable/intermittent volumetric flow rate into the batch DF/UF operation is assumed (i.e. specified amount of product collected and then batch processed) • While it is not a requirement for the inlet product concentration to be constant to the batch DF/UF step, it is assumed to be constant from the previous unit operation (i.e. single-pass tangential flow filtration). *It is assumed that a batch pool tank is used for lot definition.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Single-pass tangential flow filtration	Volumetric flow rate (L/h)	Constant	None	Volumetric flow rate (L/h)	Constant	Inline diafiltration
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme	1						
Notes	• Constant volumetric flow rate out of the single-pass tangential flow filter is assumed based on the assumption that the inlet concentration is constant • Constant product concentration out of the single-pass tangential flow filter is assumed • Constant volumetric flow rate into the in-line diafiltration unit operation is assumed to enable consistent buffer exchange • Constant product concentration into in-line diafiltration is assumed.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	DF/UF (optional) (batch)	Volumetric flow rate (L/h)	Variable/intermittent	None	Volumetric flow rate (L/h)	Variable	Final filtration and bulk fill (batch)	
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant		
Input/output pairing	B				2			
Conserved control scheme	1							
Notes	• The volumetric flow rate out of the batch DF/UF operation is variable/intermittent as it is performed in batch mode • The product concentration out of the batch DF/UF operation is constant • The volumetric flow rate into the batch final filtration and bulk fill may be variable/intermittent depending on the setup • The product concentration into the final filtration and bulk fill is constant • No intermediate vessel is assumed as the material is batch pooled in the prior unit operation and the retentate vessel acts as the feed for the bulk fill.							

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	In-line diafiltration	Volumetric flow rate (L/h)	Constant	Batch pool*	Volumetric flow rate (L/h)	Variable	UF—optional (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				2		
Conserved control scheme				3*			
Notes	• Constant volumetric flow rate out of the in-line diafiltration unit is assumed based on the assumption that the inlet concentration is constant • Constant product concentration out of the in-line diafiltration unit is assumed • Variable/intermittent volumetric flow rate into the batch UF operation is assumed (i.e. specified amount of product collected and then batch processed) • While it is not a requirement for the inlet product concentration to be constant to the batch UF step, it is assumed to be constant from the previous unit operation (i.e. in-line diafiltration). *It is assumed that a batch pool tank is used for lot definition.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	Inline diafiltration	Volumetric flow rate (L/h)	Constant	None	Volumetric flow rate (L/h)	Constant	Single-pass tangential flow filtration (optional)	
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant		
Input/output pairing	D				4			
Conserved control scheme	1							
Notes	• Constant volumetric flow rate out of the in-line diafiltration unit operation is assumed if the inlet volumetric flow rate is constant • Constant product concentration into the single-pass tangential flow filter is assumed based on the assumption of constant product concentration output from the in-line diafiltration • No surge vessel is required to accommodate the unit operation interfaces in terms of flow.							

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	UF—optional (batch)	Volumetric flow rate (L/h)	Variable	None	Volumetric flow rate (L/h)	Variable	Final filtration and bulk fill (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	B				2		
Conserved control scheme	1						
Notes	• The volumetric flow rate out of the batch UF operation is variable/intermittent as it is performed in batch mode • The product concentration out of the batch UF operation is constant • The volumetric flow rate into the batch final filtration and bulk fill may be variable/intermittent depending on the setup • The product concentration into the final filtration and bulk fill is constant • No intermediate vessel is assumed as the material is batch pooled in the prior unit operation and the retentate vessel acts as the feed for the bulk fill.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2
	Single-pass tangential flow filtration (optional)	Volumetric flow rate (L/h)	Constant	Batch pool*	Volumetric flow rate (L/h)	Constant	Final filtration and bulk fill (batch)
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant	
Input/output pairing	D				4		
Conserved control scheme				3*			
Notes	• Constant volumetric flow rate out of the single-pass tangential flow filter is assumed to ensure a consistent outlet concentration from the single-pass tangential flow filter • Constant product concentration out of the single-pass tangential flow filter is assumed • The volumetric flow rate into the batch final filtration and bulk fill is assumed be constant. However, an intermittent setup may be considered, requiring a different intermediate vessel and control scheme • The product concentration into the final filtration and bulk fill is constant. *It is assumed that a batch pool tank is used for lot definition.						

	Unit operation 1	Output		Vessel	Input		Unit operation 2	
	Single-pass tangential flow filtration (optional)	Volumetric flow rate (L/h)	Constant	None	Volumetric flow rate (L/h)	Constant	Final filtration and bulk fill (continuous)	
		Product concentration (g/L)	Constant		Product concentration (g/L)	Constant		
Input/output pairing	D				4			
Conserved control scheme	1							
Notes	• Constant volumetric flow rate out of the single-pass tangential flow filter is assumed • Constant product concentration out of the single-pass tangential flow filter is assumed • The volumetric flow rate into the continuous final filtration and bulk fill is constant • The product concentration into the final filtration and bulk fill is constant • No vessel is assumed to be needed as lot definition would be post filtration.							

Acronyms

Term	Definition
B/E chromo	Bind/elute chromatography
CLARA	Closure analysis risk assessment
CMCC	Continuous multi-column chromatography
CPA	Critical process attribute
CPP	Critical process parameter
CQA	Critical quality attribute
DACs	Disposable aseptic connectors
DF	Diafiltration
F/T chromo	Flow through chromatography
ICH	International Conference on Harmonization
ILDF	In-line diafiltration
KPA	Key process attribute
KPP	Key process parameter

Term	Definition
LpH VI	Low pH viral inactivation
mAb	Monoclonal antibody
NFF	Normal flow filter
NOR	Normal operating range
PAT	Process analytical technology
PAR	Proven acceptable range
POL	Process orchestration layer
QTPP	Quality therapeutic product parameter
pCPP	Potentially critical process parameter
pKPP	Potentially key process parameter
SPTFF	Single pass tangential flow filtration
TFF	Tangential flow filtration
UF	Ultrafiltration

References

- 1 Heise, C., Brower, M., Teitz, S., Arnold, L., Lu, R., Poggel, M., . . . Jain, S. *Biomanufacturing Technology Roadmap: Continuous Downstream Processing for Biomanufacturing An Industry Review*. BioPhorum.com. [Online] 2019. <https://www.biophorum.com/download/continuous-downstream-processing-for-biomanufacturing-an-industry-review/>.
- 2 NIIMBL. NIIMBL N-mAb Case Study. NIIMBL. [Online] 2022. <https://niimbl.force.com/s/n-mab>.
- 3 Schaefer, G., . . . (2022). *A case study to support development and adoption of integrated continuous bioprocesses for monoclonal antibodies*. NIIMBL.com. [Online] 2022. <https://niimbl.force.com/s/n-mab>.
- 4 Coffman, J., Brower, M., Connell-Crowley, L., Deldari, S., Farid, S. S., Horowski, B., Rose, S. 4, s.l. *A common framework for integrated and continuous biomanufacturing*. *Biotechnology and Bioengineering*, 118(4), Vol. 118, pp. 1735-1749.
- 5 Pleitt, K., Kozaili, J., Huber, C., Heise, C., Hussain, S., Ghodbane, M., Reeder, A., Randolph, P., Brower, M., Wiendahl, M., Collins, M., Schofield, M., Kane, B., Erdenberger, T., Moody, G. *RISK-BASED BLUEPRINT FOR PROCESS CONTROL OF CONTINUOUS BIOPROCESSING: PHASE 1: GENERIC RISK ASSESSMENT AND PROCESS CONTROL PARAMETERS*. BioPhorum.com. [Online] 2021. <https://www.biophorum.com/download/a-risk-based-blueprint-for-process-control-of-continuous-bioprocessing/>
- 7 Hernandez, Randi. *To surge or not to surge*. 16, s.l. : Genetic Engineering and Biotechnology News, 2017, Vol. 37.
- 8 Garima Thakur, Saxena Nikita , Anamika Tiwari , Anurag S Rathore. s.l. *Control of surge tanks for continuous manufacturing of monoclonal antibodies*. *Biotechnol Bioeng*, 2021, Vols. 118(5):1913-1931.
- 9 ICH. ICH Q9(R1) Guideline-Quality Risk Management. <https://www.ema.europa.eu>. [Online] Dec 21, 2021. <https://www.ema.europa.eu/en/ich-q9-quality-risk-management-scientific-guideline>.
- 10 ICH guideline Q8 (R2) on pharmaceutical development. [Online] June 22, 2017. <https://www.ema.europa.eu/en/ich-q8-r2-pharmaceutical-development-scientific-guideline>.
- 11 Rathore, Anurag S. *Continuous Processing for Production of Biotech Therapeutics*. 69 (3), s.l. : PDA Journal of Pharmaceutical Science and Technology, May 2015, Vols. May 2015, 69 (3) 333; DOI: <https://doi.org/10.5731/pdajpst.2015.01072>, p. 333.
- 12 Eissa, Mostafa. *Bioburden Control in the Biopharmaceutical Industry*. 9, s.l. : BioPharm International, 2017, Vol. 30, pp. 24-27.
- 13 Hughes, Patricia. *FDA Expectations Regarding Bioburden Control in Biotech Processes*. [Online] Sept 23, 2010. <https://vdocuments.site/bioburden-fda-hughes-patricia.html?page=10>.
- 14 *Continuous downstream processing for biomanufacturing: an industry review*. BioPhorum.com. [Online] 2019. <https://www.biophorum.com/biomanufacturing-technology-roadmap/>.
- 15 BioPhorum. *Continuous process verification: An industry position paper with an example plan*. [Online] <https://www.biophorum.com/download/cvp-case-study-interactive-version/>.
- 16 ICH. ICH Q8-Q11: *Development and Manufacture of Drug Substances (Chemical Entities and Biotechnological/Biological Entities)*. [Online] 2012. https://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/Guidelines/Quality/Q11/Q11_Step_4.pdf.

Permission to use

The contents of this report may be used unaltered as long as the copyright is acknowledged appropriately with correct source citation, as follows 'Entity, Author(s), Editor, Title, Location: Year'

<https://doi.org/10.46220/2023TS004>

Disclaimer

This document represents a consensus view, and as such it does not represent fully the internal policies of the contributing companies.

Neither BioPhorum nor any of the contributing companies accept any liability to any person arising from their use of this document.

The views and opinions contained herein are that of the individual authors and should not be attributed to the authors' employers.

CONNECT COLLABORATE ACCELERATE
is a trademark of BioPhorum Operations Group.

